Bangladesh Army University of Engineering & Technology
Department of Electrical & Electronic Engineering

Lecture: 12

Course Code: EEE 1261

Course Title: Electronic Devices and Circuits

Course Teacher: Ferdous Farjana

Designation: Lecturer, Dept. of EEE, BAUET

Reference Books

- Electronic Devices and Circuit Theory: Robert L. Boylestad and Louis Nashelsky.
- Electronic Principles: Albert P. Malvino.
- Micro Electronics Circuits: Adel S. Sedra & Keneth C. Smith: Oxford University Press.
- Operational Amplifiers and Linear Integrated Circuits: Robert F. Coughlin: Prentice Hall of India Private Limited.

2.8 CLIPPERS

The previous section on rectification gives clear evidence that diodes can be used to change the appearance of an applied waveform. This section on clippers and the next on clampers will expand on the wave-shaping abilities of diodes.

Clippers are networks that employ diodes to "clip" away a portion of an input signal without distorting the remaining part of the applied waveform.

The half-wave rectifier of Section 2.6 is an example of the simplest form of diode clipper—one resistor and a diode. Depending on the orientation of the diode, the positive or negative region of the applied signal is "clipped" off.

There are two general categories of clippers: *series* and *parallel.* The series configuration is defined as one where the diode is in series with the load, whereas the parallel variety has the diode in a branch parallel to the load.

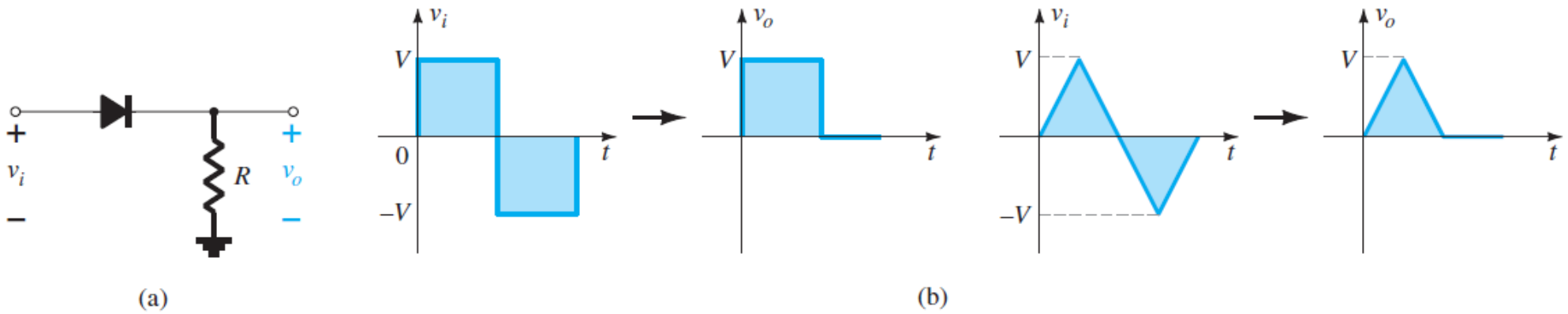


FIG. 2.68
Series clipper.

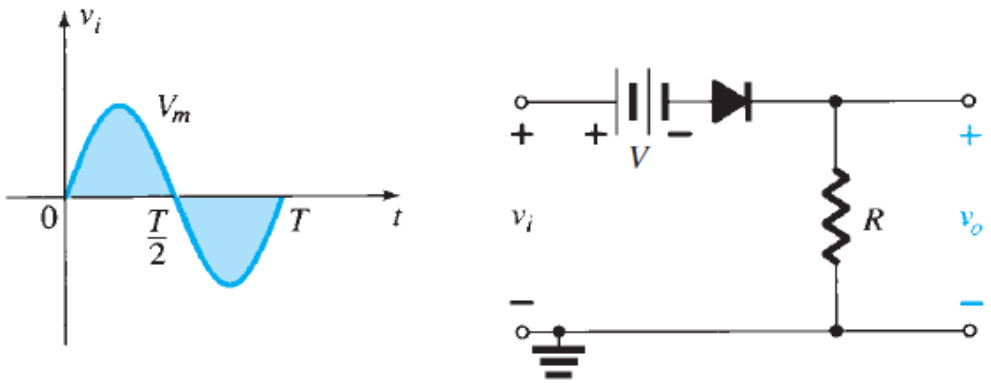


FIG. 2.69
Series clipper with a dc supply.

The addition of a dc supply to the network as shown in Fig. 2.69 can have a pronounced effect on the analysis of the series clipper configuration. The response is not as obvious because the dc supply can aid or work against the source voltage, and the dc supply can be in the leg between the supply and output or in the branch parallel to the output.

There is no general procedure for analyzing networks such as the type in Fig. 2.69, but there are some things one can do to give the analysis some direction.

First and most important:

1. **Take careful note of where the output voltage is defined.**

   In Fig. 2.69 it is directly across the resistor R. In some cases it may be across a combination of series elements.

   Next:

2. **Try to develop an overall sense of the response by simply noting the "pressure" established by each supply and the effect it will have on the conventional current direction through the diode.**

   In Fig. 2.69, for instance, any positive voltage of the supply will try to turn the diode on by establishing a conventional current through the diode that matches the arrow in the diode symbol. However, the added dc supply V will oppose that applied voltage and try to keep the diode in the "off" state. The result is that any supply voltage greater than V volts will turn the diode on and conduction can be established through the load resistor. Keep in mind that we are dealing with an ideal diode for the moment, so the turn-on voltage is simply 0 V. In general, therefore, for the network of Fig. 2.69 we can conclude that the

diode will be on for any voltage v_i that is greater than V volts and off for any lesser voltage. For the "off" condition, the output would be 0 V due to the lack of current, and for the "on" condition it would simply be $v_o = v_i - V$ as determined by Kirchhoff's voltage law.

3. Determine the applied voltage (transition voltage) that will result in a change of state for the diode from the "off" to the "on" state.

This step will help to define a region of the applied voltage when the diode is on and when it is off. On the characteristics of an ideal diode this will occur when $V_D = 0$ V and $I_D = 0$ mA. For the approximate equivalent this is determined by finding the applied voltage when the diode has a drop of 0.7 V across it (for silicon) and $I_D = 0$ mA.

This exercise was applied to the network of Fig. 2.69 as shown in Fig. 2.70. Note the substitution of the short-circuit equivalent for the diode and the fact that the voltage across the resistor is 0 V because the diode current is 0 mA. The result is $v_i - V = 0$, and so

$$\boxed{v_i = V} \tag{2.14}$$

is the transition voltage.

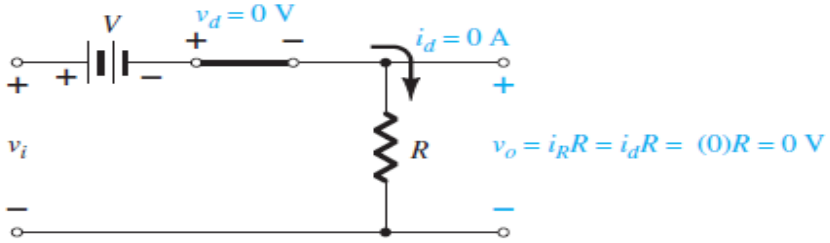


FIG. 2.70
Determining the transition level for the circuit of Fig. 2.69.

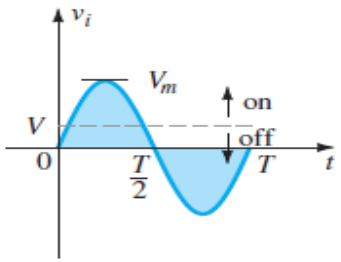


FIG. 2.71
Using the transition voltage to define the "on" and "off" regions.

This permits drawing a line on the sinusoidal supply voltage as shown in Fig. 2.71 to define the regions where the diode is on and off.

For the "on" region, as shown in Fig. 2.72, the diode is replaced by a short-circuit equivalent, and the output voltage is defined by

$$v_o = v_i - V \tag{2.15}$$

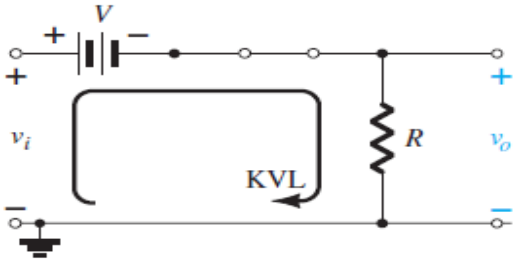


FIG. 2.72
Determining v_o for the diode in the "on" state.

For the "off" region, the diode is an open circuit, $I_D = 0$ mA, and the output voltage is

$$v_o = 0 \text{ V}$$

4. It is often helpful to draw the output waveform directly below the applied voltage using the same scales for the horizontal axis and the vertical axis.

Using this last piece of information, we can establish the 0-V level on the plot of Fig. 2.73 for the region indicated. For the "on" condition, Eq. (2.15) can be used to find the output voltage when the applied voltage has its peak value:

$$v_{o_{\text{peak}}} = V_m - V$$

and this can be added to the plot of Fig. 2.73. It is then simple to fill in the missing section of the output curve.

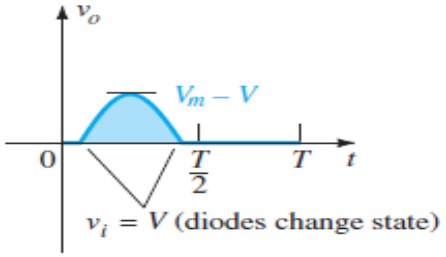


FIG. 2.73
Sketching the waveform of v_o using the results obtained for v_o above and below the transition level.

Parallel

The network of Fig. 2.81 is the simplest of parallel diode configurations with the output for the same inputs of Fig. 2.68. The analysis of parallel configurations is very similar to that applied to series configurations, as demonstrated in the next example.

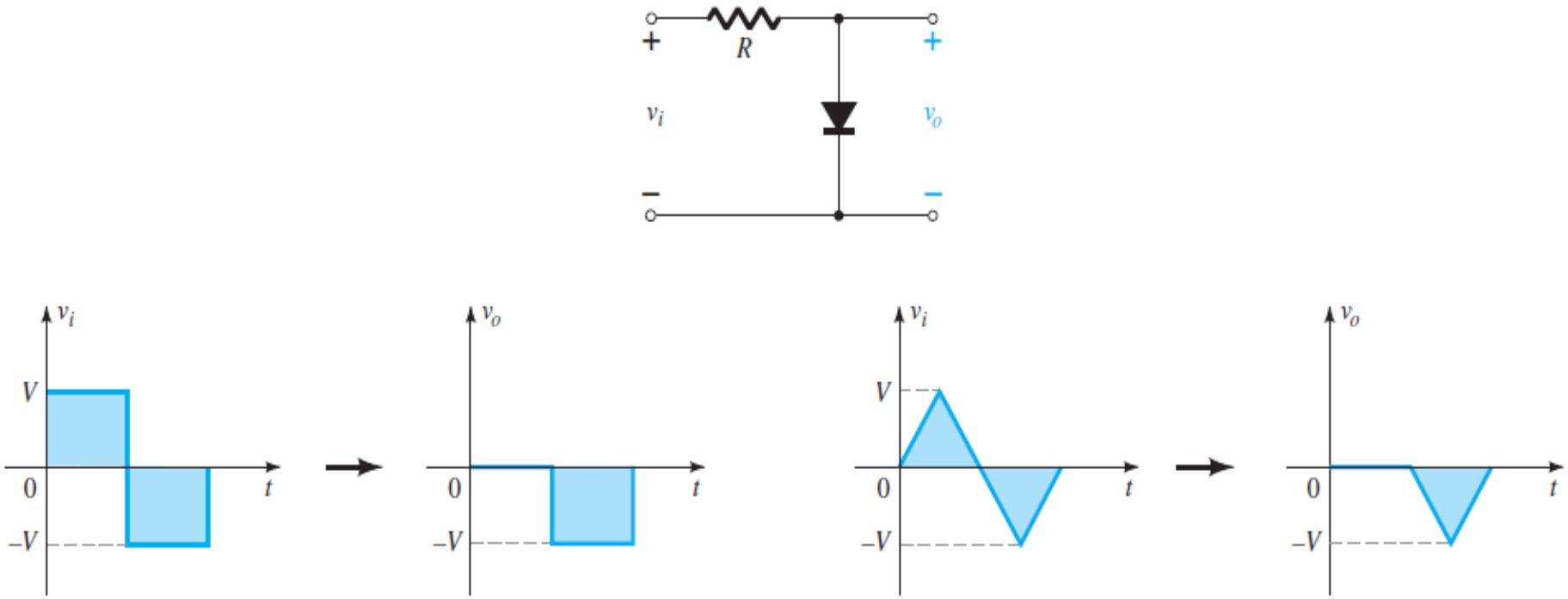
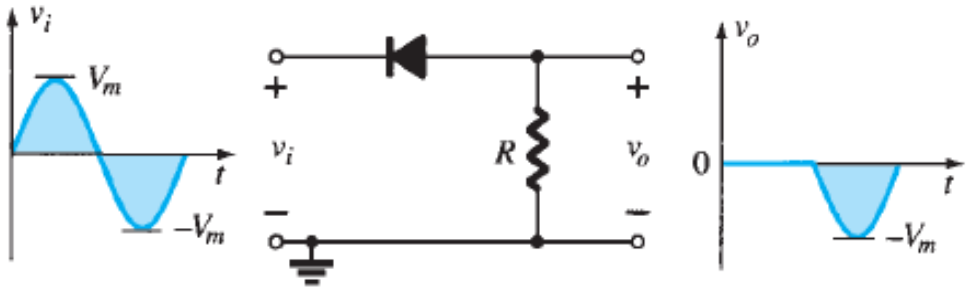


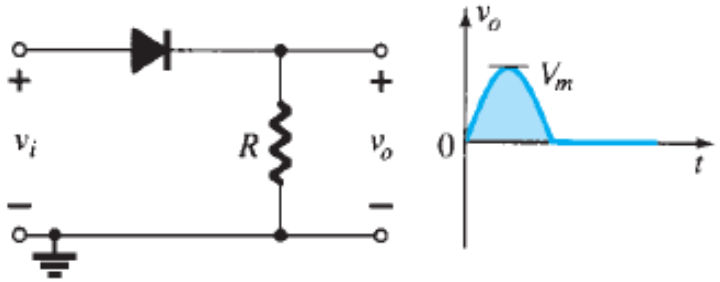
FIG. 2.81
Response to a parallel clipper.

Simple Series Clippers (Ideal Diodes)

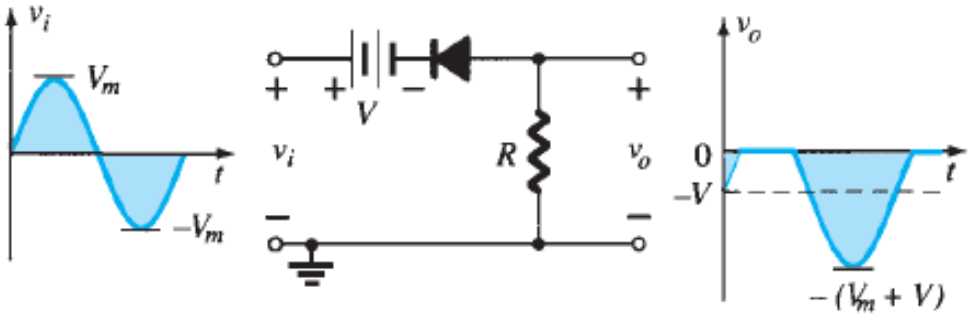
POSITIVE

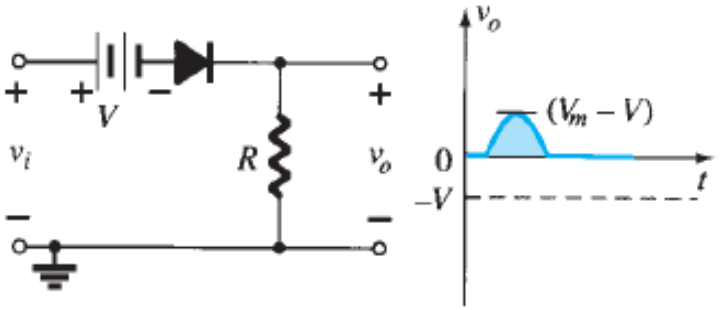


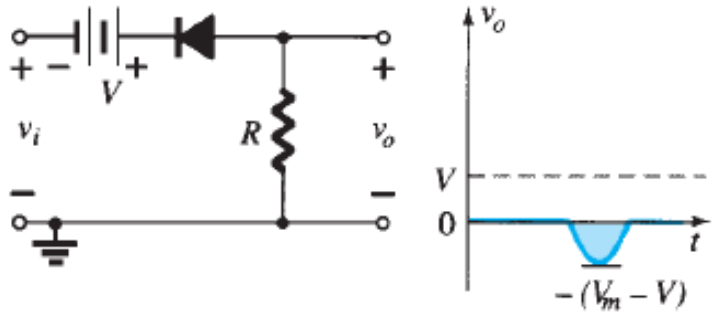
NEGATIVE

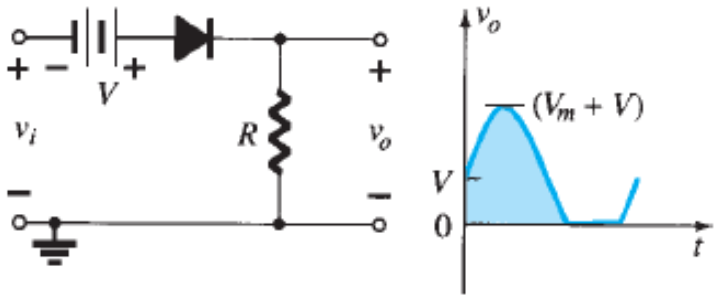


Biased Series Clippers (Ideal Diodes)

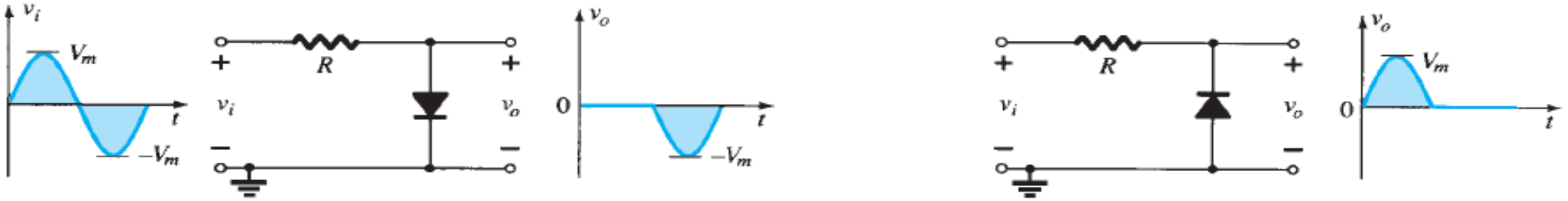








Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

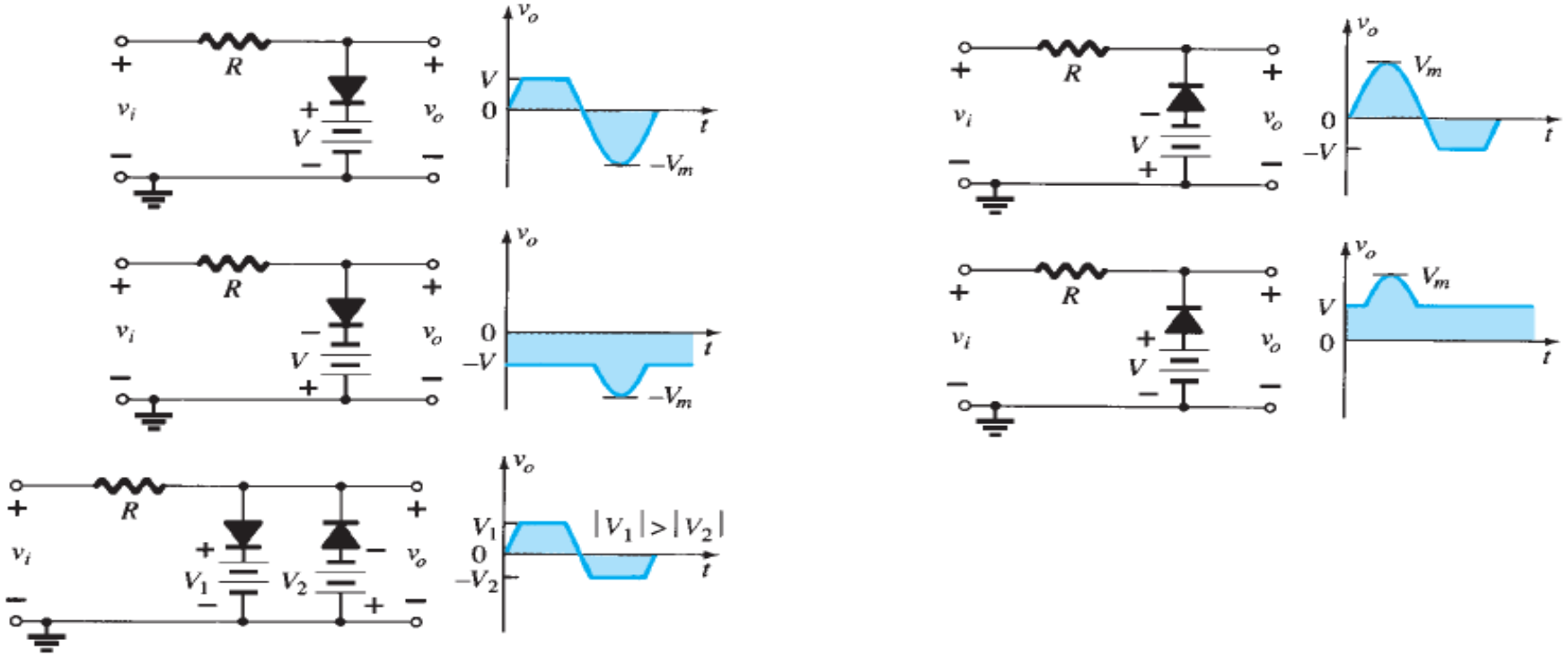


FIG. 2.88
Clipping circuits.

Summary

A variety of series and parallel clippers with the resulting output for the sinusoidal input are provided in Fig. 2.88. In particular, note the response of the last configuration, with its ability to clip off a positive and a negative section as determined by the magnitude of the dc supplies.

THANK YOU